

# Functionalized Boron Nitride-Based Modification Layer as Ion Regulator Toward Stable Lithium Anode at High Current Densities

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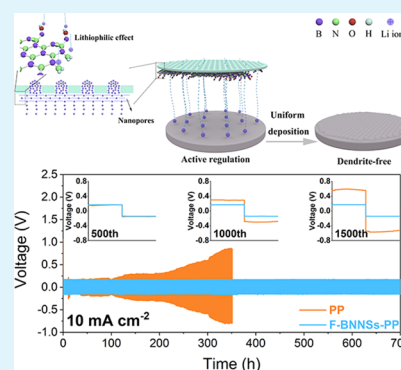
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Supporting Information

**ABSTRACT:** It is difficult to achieve higher energy density with the existing system of lithium (Li)-ion batteries. As a powerful candidate, Li metal batteries are in the renaissance. Unfortunately, the uncontrolled growth process of Li dendrites has limited their actual application. Hence, inhibiting the formation and spread of Li dendrites has become an enormous challenge. Herein, a novel composite separator is developed with functionalized boron nitride nanosheet modification layer as a Li-ion regulator to regulate Li-ion fluxes. The composite separator contains abundant polar groups and nanoscale channels and could achieve uniform electrochemical deposition via the lithiophilic effect and shunting action. Under the synergy influence of the lithiophilic effect and shunting action, Li dendrites are effectively suppressed. As proof, the Li||Li symmetrical cells with composite separators can circulate steadily for a long time under high current densities ( $10 \text{ mA cm}^{-2}$ , 800 h). Moreover, the  $\text{LiFePO}_4$ ||Li full cells display excellent long cycling performance (82% retention after 800 cycles).

**KEYWORDS:** lithium metal batteries, lithium dendrites, ion regulator, separator modification, hexagonal boron nitride, light modification layer



## 1. INTRODUCTION

Lithium (Li) metal, as a powerful candidate for the anode of Li secondary (rechargeable) batteries, possesses the highest specific capacity ( $3860 \text{ mAh g}^{-1}$ ) and the lowest redox potential ( $-3.04 \text{ V}$  vs the standard hydrogen electrode),<sup>1</sup> which is a pivotal part of the Li metal batteries (LMBs), such as Li–sulfur batteries, Li–air/oxygen batteries, and Li–carbon dioxide batteries.<sup>2–10</sup> Unfortunately, the presence of Li dendrites may cause short circuit and fire in LMBs, which greatly limits the practical application of LMBs. The protruding sites on the Li metal surface strengthen the local electric field,<sup>11</sup> which leads to uneven Li-ion fluxes on the Li metal surface, and eventually, the nonuniform deposition of Li ions leads to the Li dendrite growth.<sup>12</sup> Rampant Li dendrites could lead to incessant cracking and generation of the solid electrolyte interphase (SEI) layer on the surface of the Li metal, which accelerates the depletion of the electrolyte, leading to premature failure of the battery.<sup>8,13</sup> At the same time, rapidly spreading Li dendrites puncture the separator, resulting in a short circuit inside the battery,<sup>14</sup> which may cause fire or even explosion. Moreover, the isolated Li (also known as “dead Li”) is generated during the Li stripping process. As a consequence, Coulombic efficiency (CE) is greatly reduced, and the large amount of dead Li accumulated on the Li metal surface deteriorates the battery performance.<sup>15,16</sup> Therefore, inhibiting the Li dendrite formation and spread has become the key to the revival of LMBs.

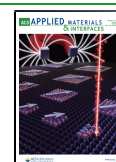
Much effective effort has been taken by the researchers to stabilize the Li metal anode: (1) Using high modulus solid or gel electrolyte to suppress dendrites,<sup>2,17–19</sup> but the low ionic conductivity greatly limits its operation at a high rate; (2) building a more stable artificial SEI is another way,<sup>20–23</sup> whereas the low strength and ionic conductivity make the interface difficult to stabilize for a long time; and (3) constructing a three-dimensional (3D) host for Li metal can lower the local current density and inhibit the expansion of Li deposition,<sup>7,24</sup> but the high dosage of electrolyte and the huge weight of the 3D host greatly reduce the energy density of the battery.

As an inactive and critical part of the battery, the separator can ensure effective ion transport and isolate the electrons.<sup>25,26</sup> Modification of the separator is an effective strategy for improving the battery performance and safety. Recently, a large number of two-dimensional (2D) materials have been used in separator modification;<sup>10,27</sup> for example, nitrogen- and sulfur-doped graphene (NSG) nanosheets were constructed on the commercial diaphragm. The defects in the NSG layer can form

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homogeneous Li-ion fluxes on the Li metal surface, resulting in the inhibition of Li dendrite growth.<sup>27</sup> Hexagonal boron nitride (h-BN) is an excellent 2D ceramic material, which has ultrahigh Young's modulus, high chemical stability, and excellent electrical insulation, and has been widely used in the LMBs, especially in the field of composite electrolytes.<sup>28–30</sup> For example, Woo et al. reported a high-modulus h-BN nanoplatelet composite gel electrolyte. The h-BN nanoplatelets were used for the solid framework of the gel electrolyte. Benefitted from the high modulus of the composite electrolyte, the composite gel electrolytes exhibit excellent performance in Li-ion batteries.<sup>29</sup>

Compared with the surface modification on Li metal, the modification on the separator seems to be more safe and simple as it can avoid the reaction of the residual solvent with Li metal. Moreover, the whole operation process can be carried out in the atmospheric environment, thus reducing the operation difficulty and cost. Previous studies on separator modification mainly focused on improving the wettability, thermal stability, and mechanical strength of the separator.<sup>31–33</sup> However, there are few studies on improving the stability of LMBs through separator modification.

In this study, h-BN was chosen as the template, and the functionalized h-BN (F-BNNSs) was obtained by a simple ball milling method. Then, uniform and tight F-BNNSs modification layers were constructed on a commercial Celgard 2400 polypropylene (PP) separator after vacuum assembly. The modification layers have abundant polar groups, which have strong affinity to Li ions.<sup>34–36</sup> Furthermore, the composite separator shows abundant nanoscale channels via the stacking of nanofilms. As a consequence, this modification layer can be used as a Li-ion regulator to stabilize Li metal. When Li ions are transported into the modification layer, the Li ions can be effectively dispersed under the synergistic effect of the shunting action and the lithiophilic effect,<sup>4,37</sup> so that Li dendrite growth can be effectively inhibited. Even if Li dendrites are produced, the modification layer can hinder the spread of dendrites through the high-strength Young's modulus (1 TPa), thus avoiding the short circuit in the battery.<sup>38–40</sup> Under the regulation of the F-BNNSs modification layer, the symmetrical cells can circulate steadily and under high current densities (10 mA cm<sup>-2</sup>, 800 h). Moreover, the modification layer is relatively thin and light, which has little influence on the energy density of the battery. Such a functionalized boron nitride nanosheets composite PP (F-BNNSs-PP) separator is expected to promote the practical application of LMBs.

## 2. EXPERIMENTAL SECTION

**2.1. Preparation of F-BNNSs.** To prepare F-BNNSs, commercial h-BN (Jonye Tech) and urea (Aladdin) crystals were mixed into a steel milling container, and the rotation speed was controlled at 500 rpm for 20 h under argon (Ar) protection. The milled product was washed with deionized water, and then the washed materials were dispersed in deionized water by ultrasonication for 1 h. The dispersion was centrifuged three times at 11 500 rpm for 15 min to precipitate the unfunctionalized h-BN. Finally, the wet product was obtained by vacuum filtration and drying at 80 °C. For comparison, the h-BN was ball-milled under the same conditions, but no urea was added during the ball milling process. The product was named MBN.

**2.2. Fabrication of F-BNNSs-PP Separators.** The above product was dispersed in *N*-methyl pyrrolidone (NMP) (Aladdin) by ultrasonication for 2 h, and then the dispersion was centrifuged three times at 11 500 rpm for 15 min to precipitate huge F-BNNSs.

Finally, the F-BNNSs colloidal dispersion was obtained, which is very stable and can be stored for long periods. F-BNNSs-PP separators were obtained by vacuum filtration. The F-BNNSs colloidal dispersion was mixed in a poly(vinylidene fluoride) (PVDF) solution (F-BNNSs/PVDF mass ratio of 8:2). After ultrasonication for 10 min, a homogeneous mixture was obtained, and then the mixture was filtered through a glass microfiltration apparatus using a commercial separator (Celgard 2400) as a filter membrane under vacuum assistance. Finally, the F-BNNSs-PP separators were obtained after drying at 60 °C for 24 h. The average F-BNNSs loading in the F-BNNSs-PP separator was about 0.1683 mg cm<sup>-2</sup>. MBN composite PP (MBN-PP) separators were prepared following the same procedure.

**2.3. Material Characterization.** The crystal information of F-BNNSs and h-BN can be obtained by X-ray diffraction (XRD) (Rigaku MiniFlex600, Cu K $\alpha$  radiation). The microtopography of all samples was carried out by a field-emission scanning electron microscope (SEM) (JEOL JSM-7500F) and a transmission electron microscope (TEM) (JEM-2010FEF, JEOL Ltd., Japan), respectively. To determine the thickness of F-BNNSs, F-BNNSs were spin-coated on Si wafers for atomic force microscopy (AFM) (Bruker Multimode 8 with a Nanoscope V controller) measurement. The chemical components of the samples were analyzed by a Fourier transform infrared spectrometer (FT-IR) (FTS6000) and an X-ray photoelectron spectrometer (XPS) (Perkin Elmer PHI 1600 ESCA system). Contact angles were measured by a contact angle measurement device (Data physics, TBU 100).

**2.4. Battery Assembly.** All cells were assembled using 2032-type coin cells in a glovebox filled with argon gas (H<sub>2</sub>O and O<sub>2</sub> < 0.1 ppm). For symmetric cells, 150  $\mu$ m thick, 12 mm diameter Li foils were used as both anode and cathode without any treatment. The electrolyte was 1.0 M lithium bis(trifluoromethane sulfonimide) in a mixture of 1,3-dioxalane and dimethyl ether (1:1 by volume) with 1 wt % LiNO<sub>3</sub> (DodoChem), and the dosage of the electrolyte in each cell was about 30  $\mu$ L. For LFP||Li full cells, commercial LiFePO<sub>4</sub> was used as the cathode and Li foils served as the anode. Moreover, the electrolyte used for full cells was 1.0 M lithium hexafluorophosphate in a 50:50 (w/w) mixture of ethylene carbonate and diethyl carbonate (DodoChem), and about 40  $\mu$ L of electrolyte was added to each cell. The details of the preparation of the electrode are as follows: lithium iron phosphate (LiFePO<sub>4</sub>) (Chembee), Super P, and PVDF (8:1:1 w/w) were mixed in NMP, and then the mixture was coated on aluminum foils. The electrodes were dried at 60 °C for 12 h in a vacuum oven, and the active material loading of LiFePO<sub>4</sub> was around 3–3.3 mg cm<sup>-2</sup>. The commercial PP separators or F-BNNSs-PP separators served as separators in all cells. Except for the LFP||Li full cells test, all tests were based on ether electrolytes with 1 wt % LiNO<sub>3</sub> as an additive.

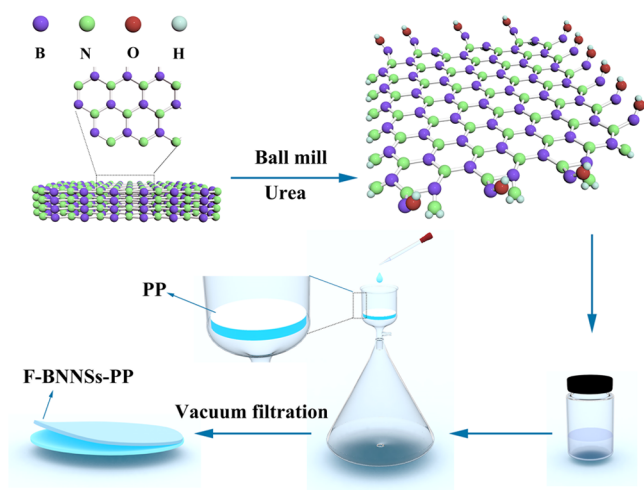
**2.5. Electrochemical Measurements.** A multichannel battery test instrument (LAND CT2001A, Wuhan LAND Electronic Co., Ltd.) was used to record galvanostatic charge/discharge processes, and the EIS tests were performed on the CHI660E (Shanghai Chenhua instrument Co., Ltd.) electrochemical workstation in a frequency range of 100 kHz to 10 mHz with an amplitude of 5 mV. The ionic conductivity of different separators was measured by symmetric coin cells and a specific value was obtained by the following formula

$$\sigma = \frac{L}{RA} \quad (1)$$

where  $\sigma$  is the ionic conductivity,  $L$  is the thickness of different separators,  $A$  is the contact area of separators and stainless steel, and  $R$  is the resistance.

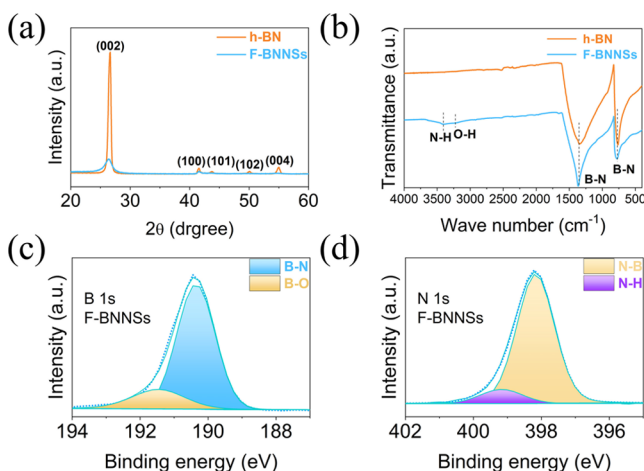
## 3. RESULTS AND DISCUSSION

The preparation process of the F-BNNSs-PP is shown in Figure 1. The F-BNNSs was obtained by a simple ball mill method, and then the F-BNNSs-PP was assembled by vacuum filtration. The size of h-BN was reduced from 1–4  $\mu$ m to 50–100 nm, while retaining its intrinsic lamellar morphology after



**Figure 1.** Schematic graph of the functionalization of h-BN and preparation of a composite separator.

ball milling (Figure S1 in the Supporting Information). The average thickness of F-BNNSs was about 5 nm as seen under AFM (Figure S2 in the Supporting Information). More crystal information can be obtained from XRD patterns; both materials show two similar characteristic peaks at 41.6 and 26.5°, which correspond to the (100) and (002) crystallographic planes (PDF 34-0421). Compared with h-BN, all characteristic diffraction peaks of F-BNNSs appear weaker and show wider peaks, indicating that the h-BN is smaller and thinner after ball milling (Figures 2a, and S3 in the Supporting



**Figure 2.** (a) XRD patterns of h-BN and F-BNNSs. (b) FTIR spectra of h-BN and F-BNNSs. XPS spectra of F-BNNSs: (c) core-level B 1s; (d) core-level N 1s.

Information).<sup>41</sup> The thickness of F-BNNSs can be obtained by Scherrer's equation

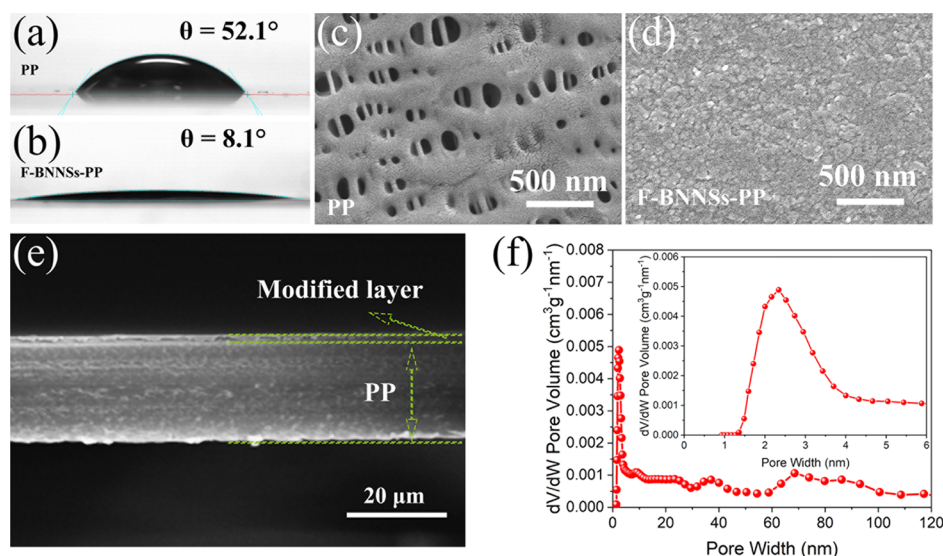
$$L_C = \frac{k\lambda}{\beta \cos \theta} \quad (2)$$

where  $\lambda$  is the wavelength of the radiation (0.154056 nm),  $\beta$  is the full-width at half maximum,  $\theta$  is the diffraction angle of (002) peak, and  $k$  is the Scherrer constant (0.89). A part of the value required for calculation can be obtained from Figure S4. The calculated thickness of F-BNNSs was 6.06 nm, which coincided with the result of AFM (Figure S2 in the Supporting

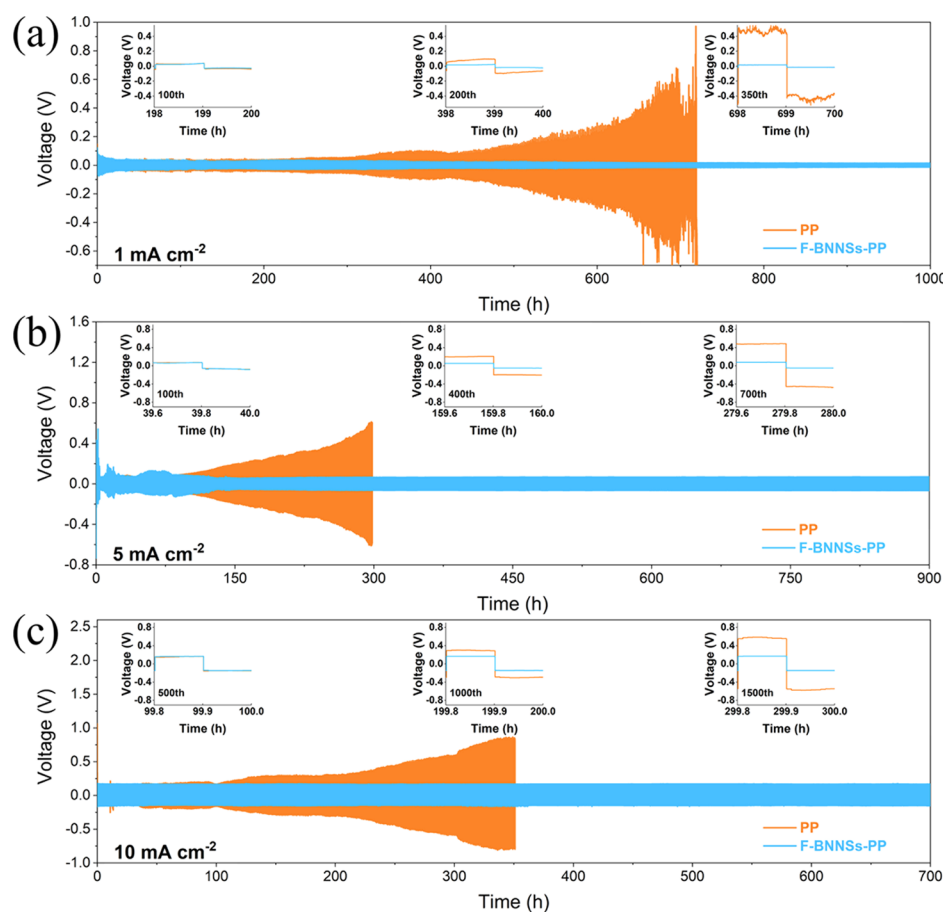
Information). More information on the functional groups contained in the material can be obtained by FT-IR spectrum and XPS. As shown in Figure 2b, h-BN exhibits two typical infrared absorption peaks at about 1350 and 770 cm<sup>-1</sup>, which can be attributed to the stretching vibrations of B-N<sup>42</sup> and the bending vibrations of B-N-B,<sup>43</sup> respectively. In contrast, except for the peaks of B-N, the stretching vibrations of O-H (3200 cm<sup>-1</sup>) and N-H (3400 cm<sup>-1</sup>) are also detected from F-BNNSs.<sup>41</sup> The B 1s spectra display two peaks at 190.4 and 191.5 eV, which can be attributed to the B-N bond and B-O bond in h-BN (Figure S5a in the Supporting Information), respectively. The latter is mainly caused by the impurities during the preparation process.<sup>44</sup> However, there are many more B atoms bound to O atoms after functionalization, which can be relevant to the increase of hydroxyl groups (Figure 2c). Figures 2d and S5b show the N-B peaks at 398.2 eV (N 1s) of F-BNNSs and h-BN, respectively. However, F-BNNSs show a N-H peak at 399.2 eV, which demonstrates that -NH<sub>2</sub> exists in F-BNNSs.<sup>45</sup> The -OH and -NH<sub>2</sub> are derived from the hydrolysis of h-BN during the washing process<sup>46</sup> and the chemical reaction between urea and h-BN,<sup>41</sup> respectively. From the above test results, we can conclude that F-BNNSs modified by -NH<sub>2</sub> and -OH were successfully prepared with the aid of urea. The FT-IR spectrum and the SEM image of MBN are shown in Figure S6. Compared with F-BNNS, MBN presents a lamellar morphology but lacks -NH<sub>2</sub> and -OH, which shows that simple ball milling cannot change the chemical composition of h-BN.

The F-BNNSs colloidal solutions were obtained by dispersing F-BNNSs with *N*-methyl pyrrolidone (NMP) by ultrasonication. It shows a typical Tyndall effect even after standing for a month indicating that F-BNNSs can be well dispersed and can maintain long-term stability in NMP. Good dispersion ensures the uniformity of the modification layer during the filtration process. In contrast, MBN dispersion did not exhibit the Tyndall effect. After 3 days, a large amount of white solid precipitate appeared at the bottom of the bottle, which indicates that MBN has poor dispersibility in NMP solution (Figure S7 in the Supporting Information). This phenomenon may be caused by the lack of polar groups in MBN. Then, the F-BNNSs modification layer was built on the commercial separator via vacuum assistance. As presented in Figure S8, the composite separator remains intact even after folding, indicating a good adhesion between the separator and the F-BNNSs layer. The poor affinity between the separator and electrolyte not only increases the interface impedance but also affects the uniformity of Li-ion fluxes, which is one of the major drawbacks of the commercial separator.<sup>47-49</sup> The contact angle between the PP separator and the electrolyte is 50.2°, and it reduces to 28.4° with the MBN-PP separator. However, the F-BNNSs separator shows better wettability to the electrolyte (Figures 3a,b and S9a in the Supporting Information), which is mainly because the polar groups in the F-BNNSs-PP separator increase the affinity between the electrolyte and the separator. The excellent wettability of the F-BNNSs-PP separator helps reduce the battery resistance and the dispersion of Li-ion fluxes.<sup>50</sup> The thickness of the F-BNNSs layer is about 1.5 μm and the composite separator surface is uniformly covered by nanomaterials; the other side of the F-BNNSs-PP separator still maintains the original morphology of the PP separator (Figure 3c-e). In contrast, the MBN separator surface exhibits an uneven modification layer (Figure S10a in the Supporting Information), which is





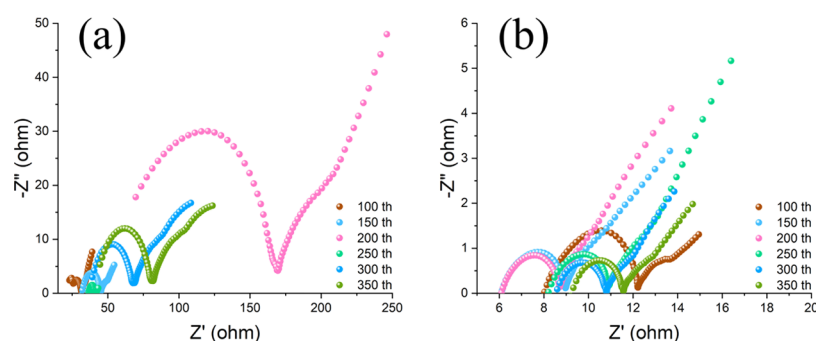
**Figure 3.** Contact angles of the electrolyte toward (a) PP and (b) F-BNNSs-PP. Top-view: SEM images of (c) the PP separator and (d) the F-BNNSs-PP separator. (e) Cross-sectional SEM image of the F-BNNSs-PP separator. (f) Pore size distribution curve of the F-BNNSs-PP separator.



**Figure 4.** (a–c) Charge and discharge curves of symmetrical cells with PP separator and F-BNNSs-PP separator at different current densities.

mainly caused by the poor dispersion of MBN in the NMP solvent (Figure S7d,e in the Supporting Information). The porous structure of the F-BNNSs-PP separator was surveyed by the  $\text{N}_2$  adsorption sorption test (Figures 3f, and S11 in the Supporting Information); typical type I isotherms reveal that the composite separator has microporous properties. In addition to macropores (50–100 nm) inherent in the

composite separator, a large number of smaller micropores (1.5–4 nm) have also been detected, which were formed by the stacking of nanosheets. Compared to the cell using PP separator, the interface impedance of the symmetrical cell is significantly reduced after using F-BNNSs-PP as a separator (Figure S9b in the Supporting Information). The decrease in electrochemical impedance can be attributed to two factors.

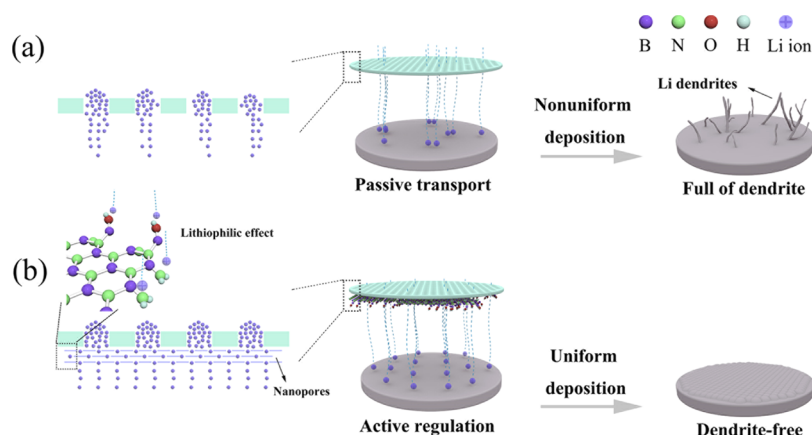


**Figure 5.** Nyquist plot of LillLi cells with (a) a PP separator and (b) an F-BNNs-PP separator after different cycles at a current density of 5 mA  $\text{cm}^{-2}$  with an areal capacity of 1 mAh  $\text{cm}^{-2}$ .

The F-BNNs-PP separator with the modification layer contains abundant polar groups, which can increase the affinity of the separator for the electrolyte (Figure 3a,b).<sup>34</sup> Moreover, the strong interaction between polar groups and Li ions can enhance the transport kinetics of Li ions at the electrode interface.<sup>51</sup> As a consequence, the symmetrical cell with an F-BNNs-PP separator displays lower interface impedance compared with a PP separator. However, the MBN-PP separator exhibits the largest interface impedance among them, which shows that the existence of polar groups is beneficial to reduce the interface impedance (Figure S9b in the Supporting Information). Coating an inorganic material on the separator can increase battery safety, but it also loses a lot of energy density. However, using F-BNNs as a coating material had only a slight effect on the weight of the original separator (Figure S12 in the Supporting Information). Besides, the shrinkage rate of the battery diaphragm at different temperatures can reflect the thermal stability of the separator. Compared to the PP separator, the F-BNNs-PP separator exhibits a lower degree of curl at high temperature, which indicates that the composite separator has better thermal stability (Figure S13 in the Supporting Information). It could greatly improve the safety performance of LMBs at increased temperature. The ionic conductivity of the PP separator is 0.160  $\text{mS cm}^{-1}$ . The ionic conductivity of the F-BNNs-PP separator (0.255  $\text{mS cm}^{-1}$ ) is slightly increased, owing to the increased wettability of the separator (Figure S14 in the Supporting Information).

To study the influence of the ion regulator on the Li dendrite growth, the LillLi symmetric cells based on PP and F-BNNs-PP separators cycling at different current densities were investigated, where the same cycling capacity of 1 mAh  $\text{cm}^{-2}$  was controlled. The cell using the F-BNNs-PP separator shows good long cycle performance, and the polarization voltage of the symmetrical cell is stable at about 12 mV after a long cycle of 1000 h. In sharp contrast, the cell using the PP separator shows worse cycle performance. Initially, the symmetric cell shows a stable overpotential similar to the symmetric cell using the F-BNNs-PP separator, whereas the overpotential of the cell has been increasing over time and it increases to 769 mV after a long cycle of 718 h (Figure 4a). However, the LillLi symmetric cell with the MBN-PP separator showed the worst performance among them, which exhibits larger voltage hysteresis and fluctuation (Figure S15 in the Supporting Information). The fluctuation of the voltage hysteresis is mainly caused by the nonuniform modification layer (Figure S10a in the Supporting Information). The uneven coating layer will lead to asymmetrical Li-ion fluxes,

which will affect the uniformity of Li deposition. The larger polarization voltage is due to the larger interface impedance. For a symmetric cell with PP separator, when a higher current density is applied, the deterioration of the cell caused by inhomogeneous deposition of Li ion is more severe and the polarization voltage surges to 500 mV when the cycle time only reaches 300 h. While using the F-BNNs-PP separator, the overpotential of the symmetric cell is only 70 mV after a long cycle of 900 h (Figure 4b). The F-BNNs-PP separator still shows excellent performance compared to the PP separator when the current density increases further and the voltage hysteresis of the cell with F-BNNs-PP separator is stable at around 170 mV even after 700 h of repeated deposition/stripping (Figure 4c). Compared with the previously reported literature, the F-BNNs-PP composite separator exhibits excellent performance in inhibiting Li dendrites (Table S1 in the Supporting Information). The stability of the interface of the cells during long cycling is evaluated by EIS. The symmetric cell with the PP separator exhibits poor interface stability and the interface impedance fluctuates greatly as the cycle progresses (Figure 5a). By comparison, the cell using F-BNNs-PP separator shows good interface stability and the interface impedance is still only 12  $\Omega$  even after 350 cycles (Figure 5b). As a powerful tool for observing the microstructure of materials, SEM was used to observe the morphology of Li deposition. For the cell using the commercial separator, a lot of mossy dendrites appeared on the Li metal surface, and many voids were created after 50 cycles. As the cycle time increases, the surface morphology of the Li metal surface deteriorates more severely (Figure S16 in the Supporting Information). By contrast, after introducing the F-BNNs-PP separator, the Li metal surface was greatly stabilized and still maintains a tight and relatively smooth morphology with the increase of cycle number, which indicates that the Li dendrite growth is inhibited to a large extent by the modification layer (Figure S17,18a in the Supporting Information). Compared with the F-BNNs-PP separator, the Li metal exhibits an uneven surface with the MBN-PP separator after cycling (Figure S18b in the Supporting Information), which may be related to the uneven coating layer. The composition and morphology of the F-BNNs-PP separator are consistent before and after cycling, which indicates that the composite separator has good mechanical and chemical stability (Figures S19 and S20 in the Supporting Information). XPS spectrum was used to study the influence of different separators on the formation of SEI layer on Li metal surfaces. The XPS test results are shown in Figures S21–S23; the composition of the Li metal surface with various separators

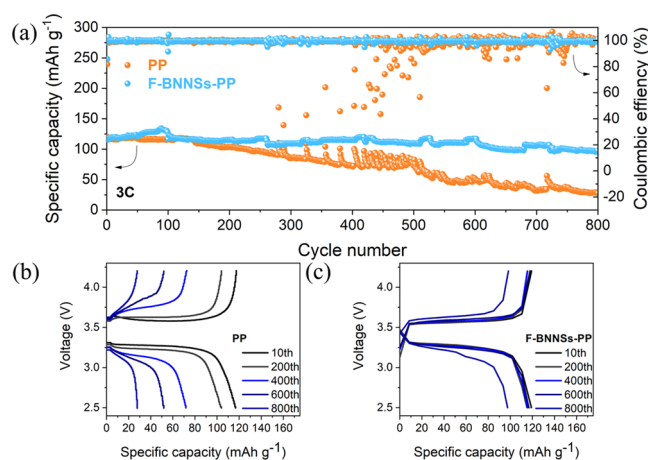


**Figure 6.** Schematic diagram of Li-ion deposition on the Li metal surface under the influence of different separators. (a) Because the PP separator has a larger pore size and a poor affinity for Li ions, the locally strengthened electric field on the Li metal surface will induce the local Li deposition, which results in the Li dendrite growth. (b) Polar groups on the F-BNNSs-PP separator exhibit a strong affinity for Li ions, and its smaller nanopores help in dispersion of Li ions, thereby effectively inhibiting Li dendrite growth.

is the same, which indicates that the F-BNNSs layer does not affect the components of the SEI layer.

A mechanism diagram was constructed to illustrate the role played by the modification layer in ion transport. The commercial PP separator is made of polypropylene, as a nonpolar compound, which has inferior affinity for Li ions. Therefore, there is a lack of affinity between Li ions and methyl groups.<sup>34</sup> Besides, the PP separator with a larger pore size will cause uneven Li-ion fluxes. Hence, when the battery uses a PP separator, uneven Li-ion fluxes easily induce local deposition behavior, resulting in a rapid spread of Li dendrites (Figure 6a), while the F-BNNSs-PP separator can actively regulate the diffusion of Li ions. The polar groups on F-BNNSs act as lithiophilic sites that disperse the Li-ion fluxes via attraction forces. Furthermore, a large number of nanoscale channels of smaller size help further disperse Li ions (shunting action). Under the synergistic effect of the shunting action and the lithiophilic effect, the Li-ion fluxes in the F-BNNSs-PP separator are more uniform. As a consequence, the growth of Li dendrites was effectively controlled (Figure 6b).

To evaluate the practical application of the F-BNNSs-PP separator, the full cells were assembled with various separators and commercial  $\text{LiFePO}_4$  served as the cathode and Li metal was used as the anode. The electrochemical properties of the cells were investigated at a cycling rate of 3C ( $1\text{C} = 170\text{ mAh g}^{-1}$ ). Both the cells with F-BNNSs-PP separator or PP separator exhibited similar initial capacities of  $120\text{ mAh g}^{-1}$ , while differences in cycling stability appeared after 130 cycles. The cycling performance of the full cell using the PP separator gradually deteriorates as the cycle progresses (Figure 7a,b). This result is mainly caused by the increased thickness of the dead Li layer and continuous consumption of the electrolyte caused by the uncontrollable Li dendrites. However, the full cell shows superior electrochemical performance when the F-BNNSs-PP separator is used, which exhibits excellent cyclic stability (a capacity retention of 82% after 800 cycles) and a high average CE of 99.5% (Figure 7a,c). This result shows that the ion regulator effectively regulates the homogeneous Li-ion fluxes during the cycles, and thus the spread of dendrites is inhibited in the full cells.



**Figure 7.** Electrochemical performance of LFP||Li cells with diverse separators. (a) Cycle performance of LFP||Li cells with various separators. Selected discharge/charge curves of LFP||Li cells with (b) a PP separator and (c) an F-BNNSs-PP separator.

#### 4. CONCLUSIONS

A novel light F-BNNSs modification layer was built on a commercial separator by simple vacuum filtration, which acts as an ion regulator by the synergistic effect of shunting action and the lithiophilic effect to regulate Li-ion fluxes. Besides, the F-BNNSs-PP displays an excellent affinity for the electrolyte, which further facilitates the uniform transport of Li ions. Under the synergy of the above effects, Li dendrites are effectively inhibited. As proof, the symmetrical cells with the F-BNNSs-PP separator can circulate steadily for a long time under high current densities ( $10\text{ mA cm}^{-2}$  800 h) and the LFP||Li full cells show superior cycling stability. It is worth mentioning that the composite separator has better thermal stability, which could greatly improve the safety performance of LMBs at increased temperature. As an essential part of the battery, our study proposed an excellent composite separator for promoting the development of LMBs.

#### ■ ASSOCIATED CONTENT

##### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.0c16354>.



SEM, TEM, and AFM image of materials; XRD pattern, XPS spectrum, and FT-IR spectrum of materials; photos of as-prepared F-BNNSs and MBN solutions; the weight, SEM images of different separators; contact angles of electrolyte toward different separators. EIS spectra of Li/Li and steel/steel symmetric cells;  $N_2$  adsorption–desorption isotherm of F-BNNSs-PP separator; photos of different separators at different temperatures; discharging/charging profiles of different separators; SEM images of Li metal surface with various separators after the cycle. FTIR spectra and SEM images of F-BNNSs-PP separator before and after cycling; XPS spectra of Li metal after cycled with different separators (PDF)

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### Author Contributions

T.M. and R.W. performed the synthesis experiment and the characterization and electrochemical measurements. S.J. and S.Z. helped in analyzing the data. L.L., J.S., and Y.C. helped in revising the manuscript. J.L. and Z.T. directed the study, discussed the results, and wrote the manuscript.

### Notes

The authors declare no competing financial interest.

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